EE 224

DIGITAL SYSTEMS

COURSE PROJECT

(Pen Paper)

Guide: Prof. Virendra Singh

TEAM MEMBERS:

Maitranya Mahesh Patil (23B1239)

Prathmesh Santosh Baral (23B1260)

Piyush Manish Gube (23B1222)

Vishnu Bijarniya (23B1251)

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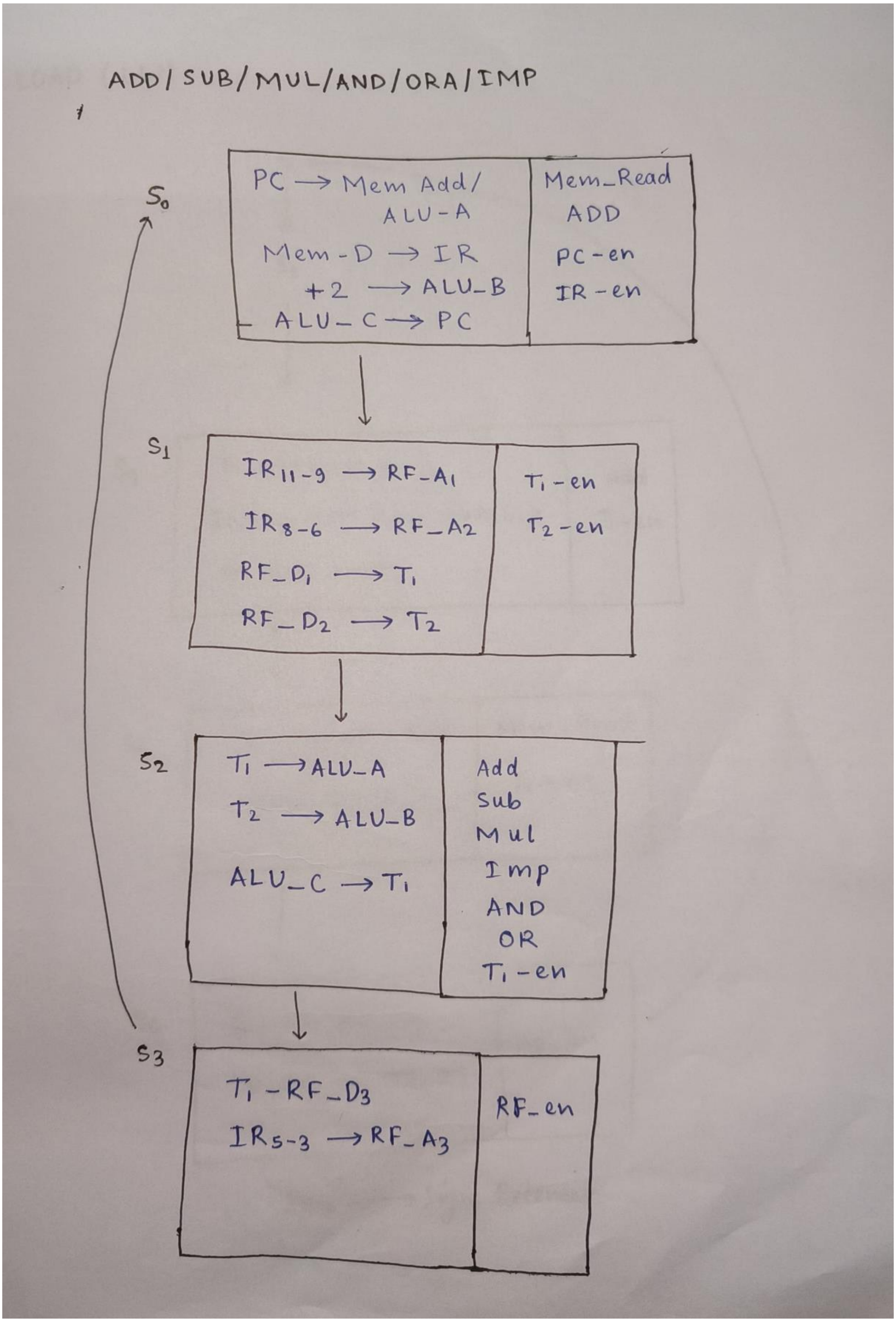
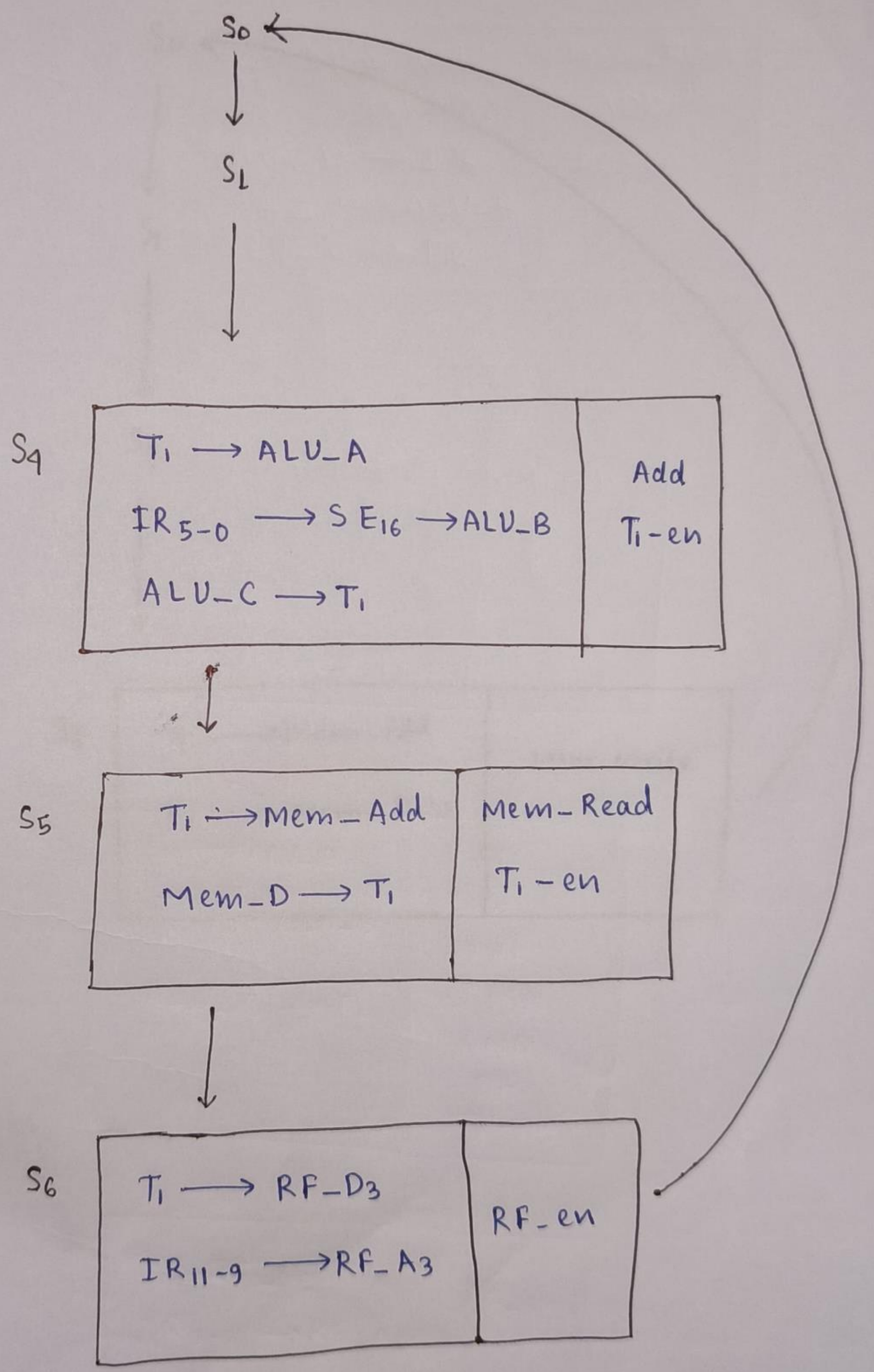
STATE DESCRIPTION

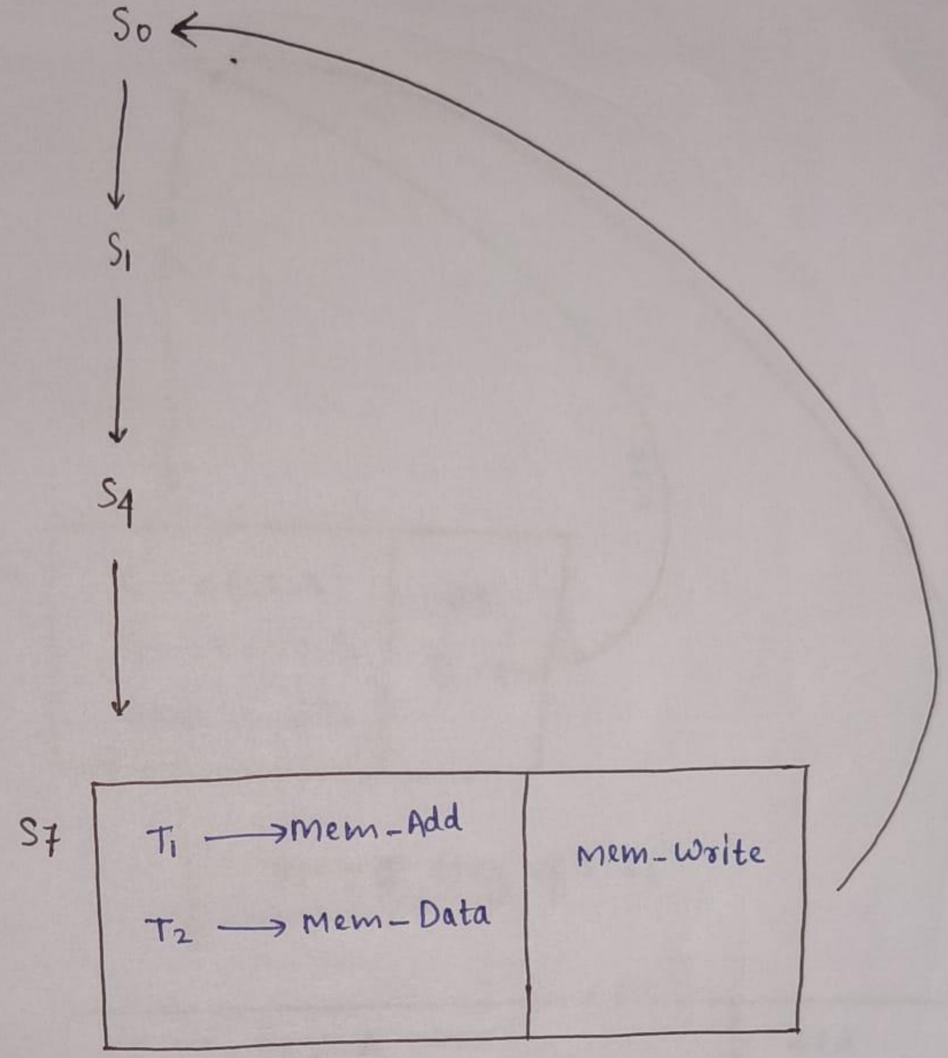


LOAD (LW)

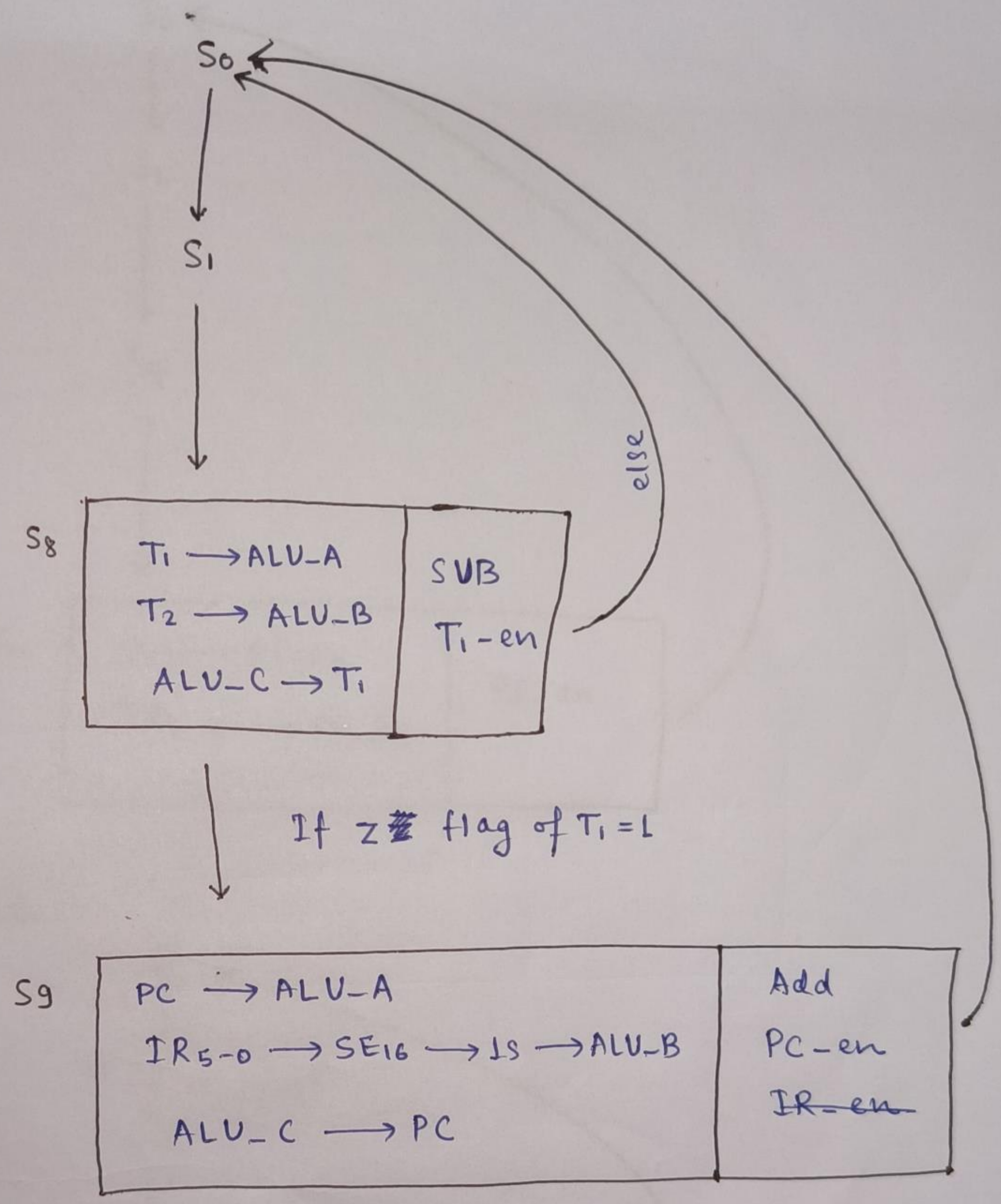


$SE16 \longrightarrow$ Sign Extension

STORE (SW)

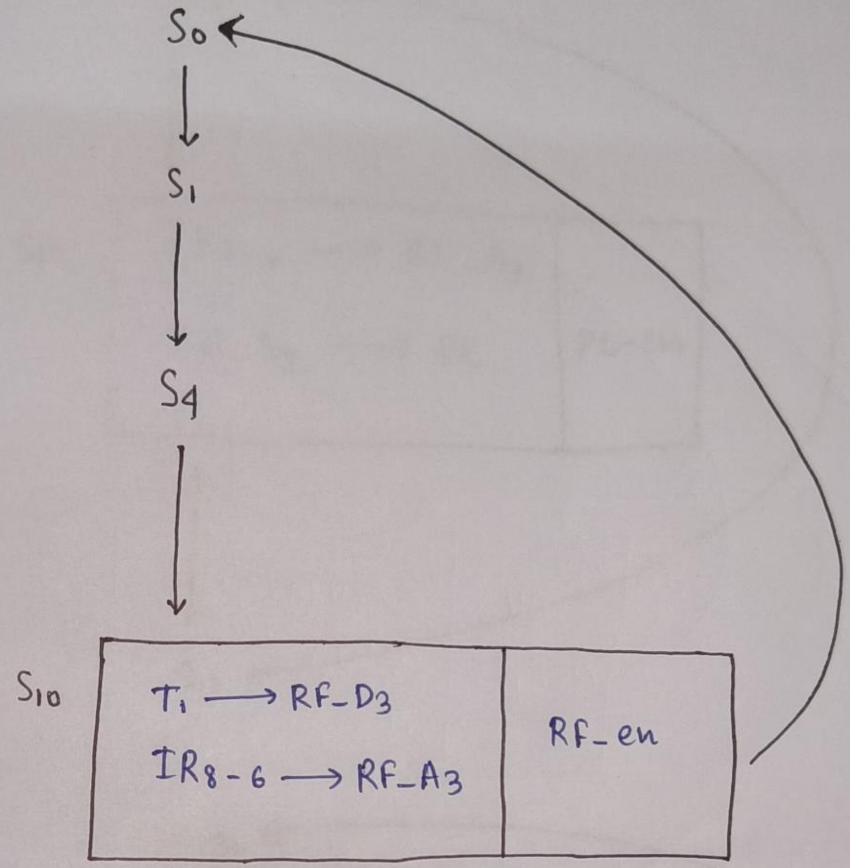


BEQ

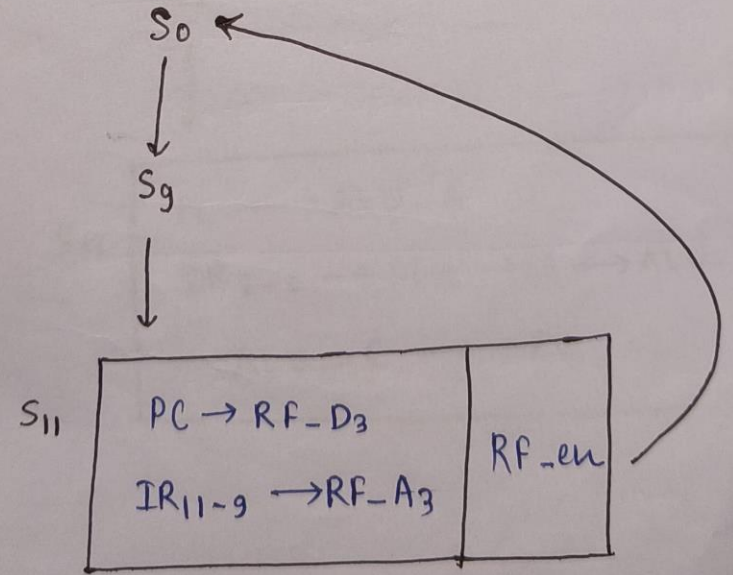


1s → 1 bit shift (to left and add 0 at last)

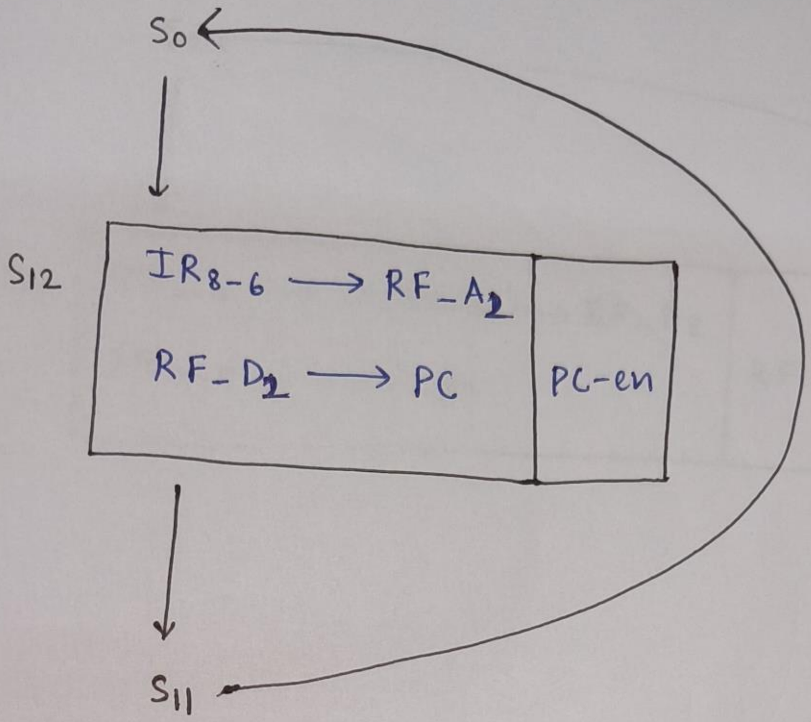
ADD Immediate



JAL

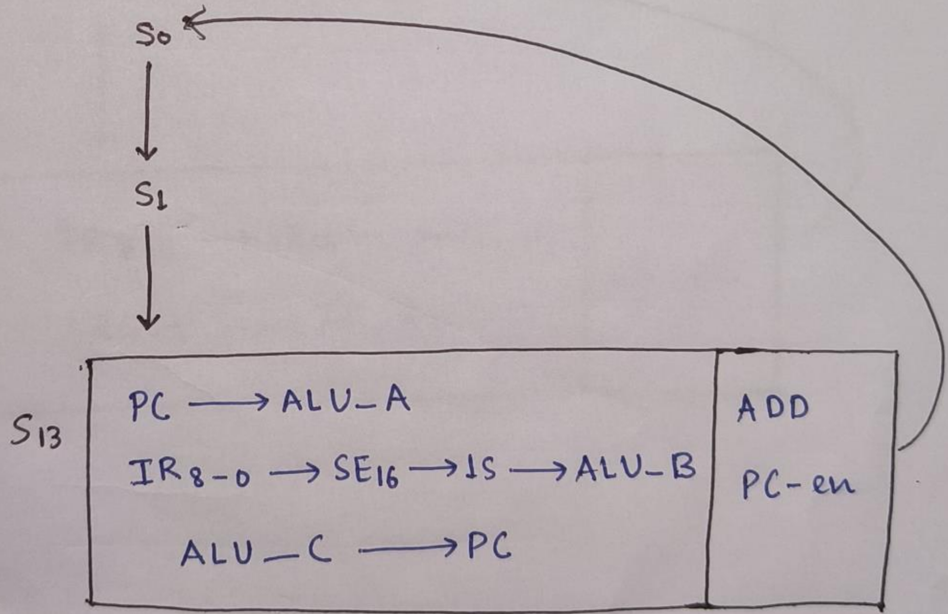


JLR



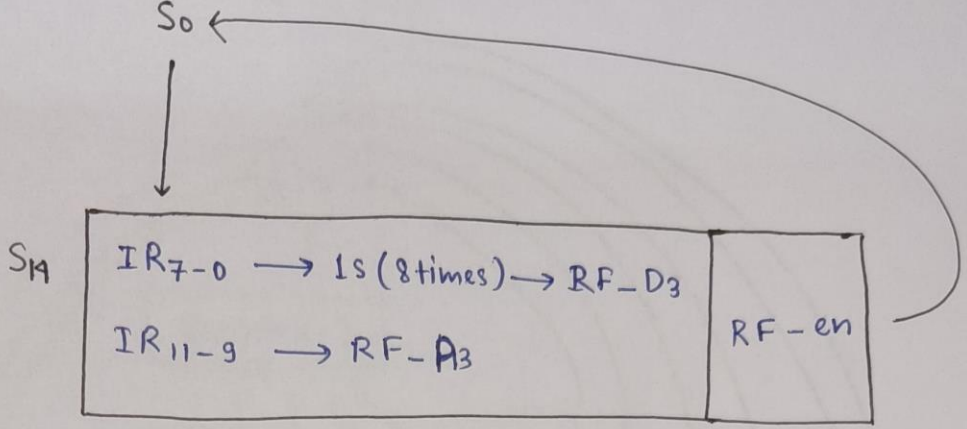
S_0
S_{12}
$IR_{8-6} \longrightarrow RF\_A_2$
$RF\_D_2 \longrightarrow PC$
PC-en
S_{11}

J



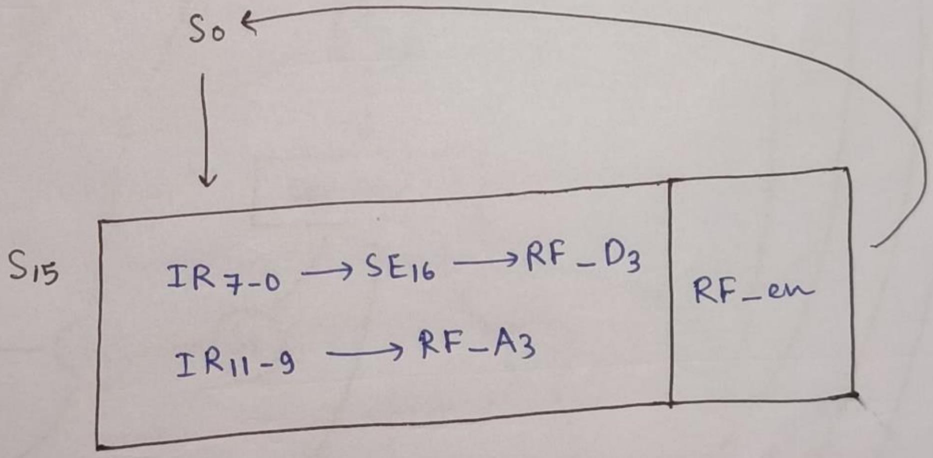
S_0
S_1
S_{13}
$PC \longrightarrow ALU\_A$
$IR_{8-0} \longrightarrow SE_{16} \longrightarrow 1S \longrightarrow ALU\_B$
$ALU\_C \longrightarrow PC$
ADD
PC-en

LHI



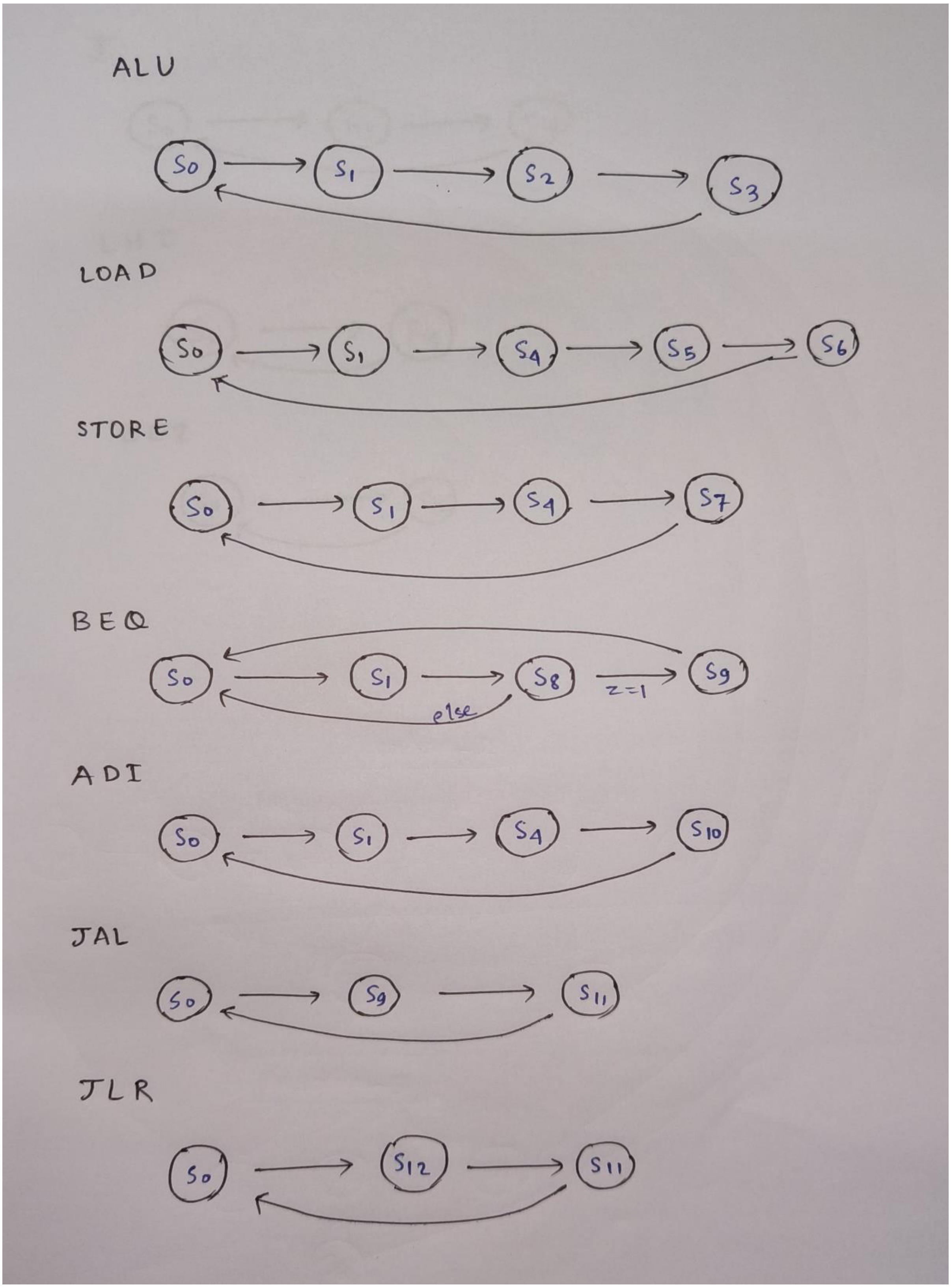
S_0
S_{14}
$IR_{7-0} \rightarrow 1s$ (8 times) $\rightarrow RF\_D_3$
$IR_{11-9} \rightarrow RF\_A_3$
RF - en

LLI

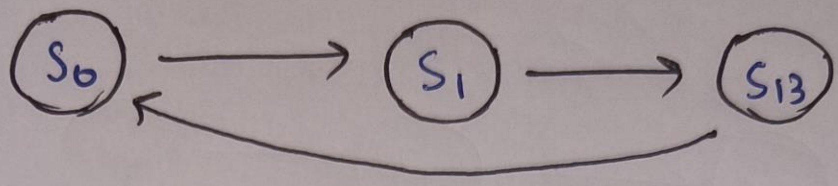


S_0
S_{15}
$IR_{7-0} \rightarrow SE_{16} \rightarrow RF\_D_3$
$IR_{11-9} \rightarrow RF\_A_3$
RF_en

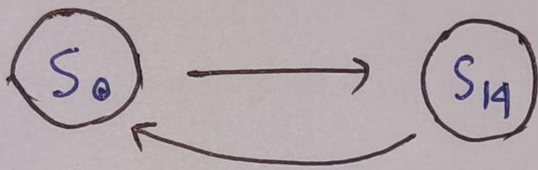
STATE FLOW


ALU
S0
S1
S2
S3
LOAD
S0
S1
S4
S5
S6
STORE
S0
S1
S4
S7
BEQ
S0
S1
S8
S9
z=1
else
ADI
S0
S1
S4
S10
JAL
S0
S9
S11
JLR
S0
S12
S11

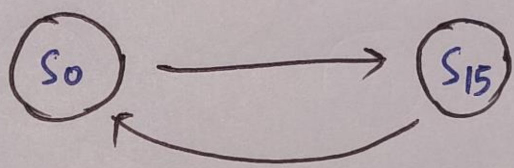
J



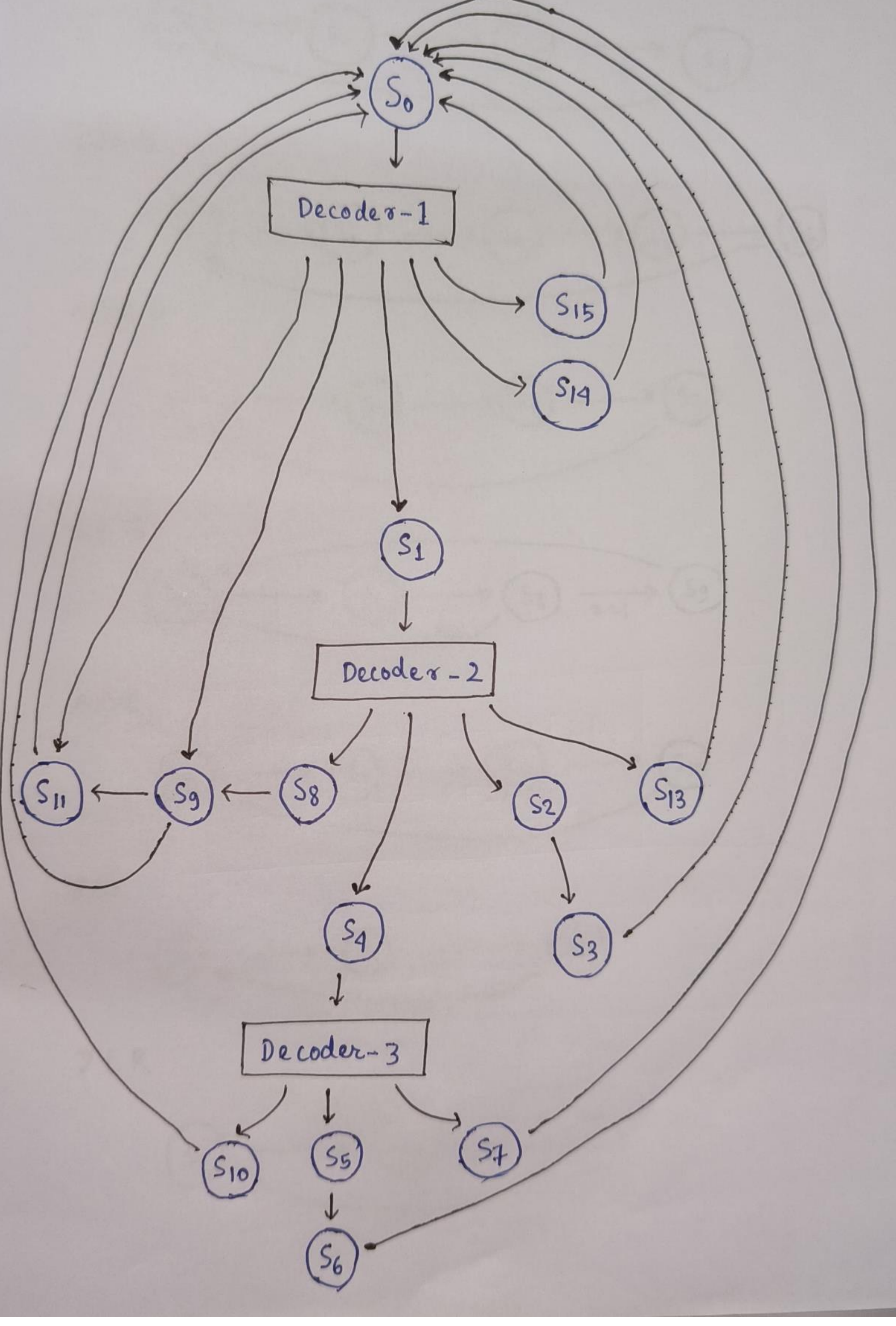
LHI



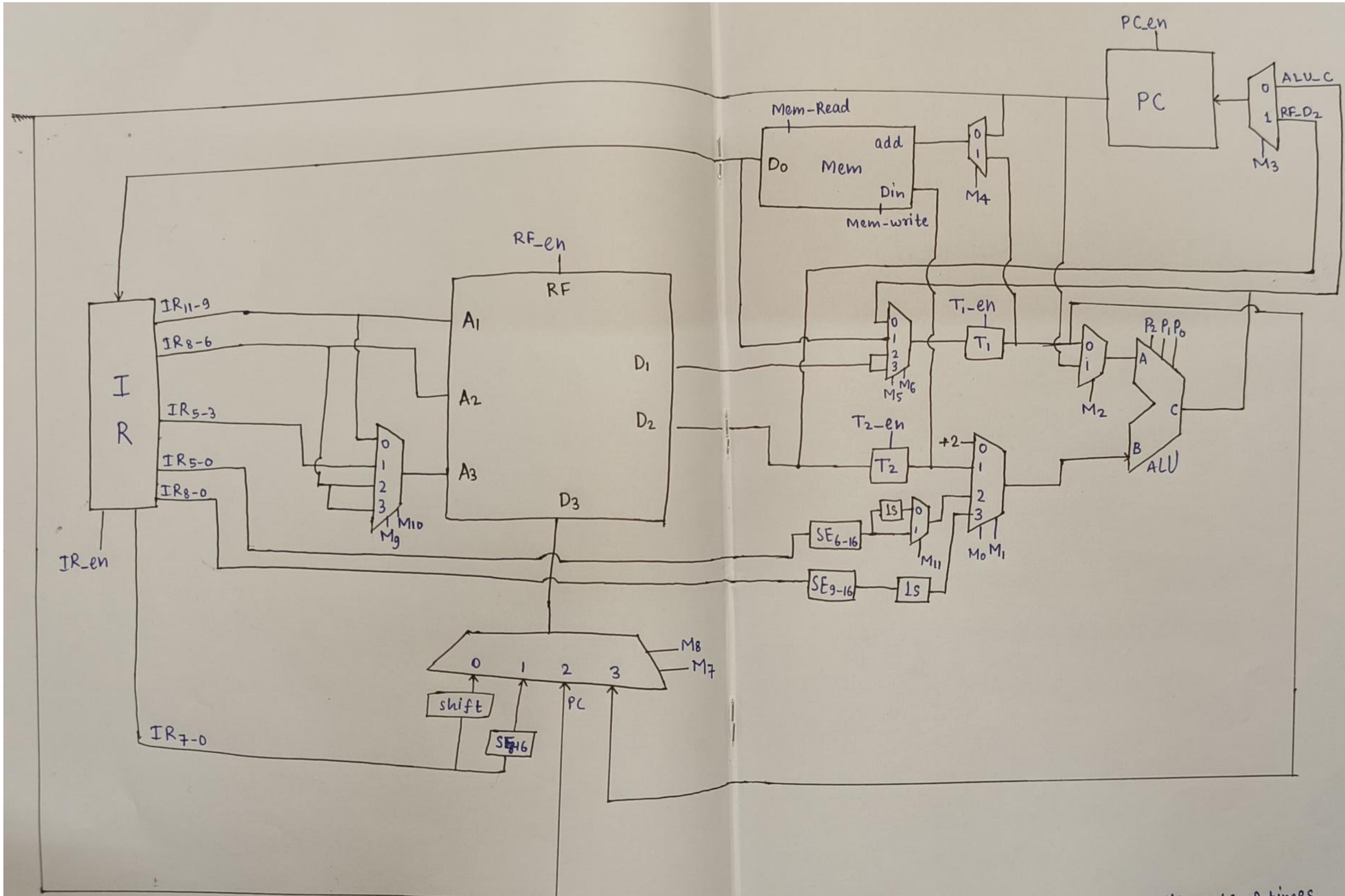
LLI



FSM

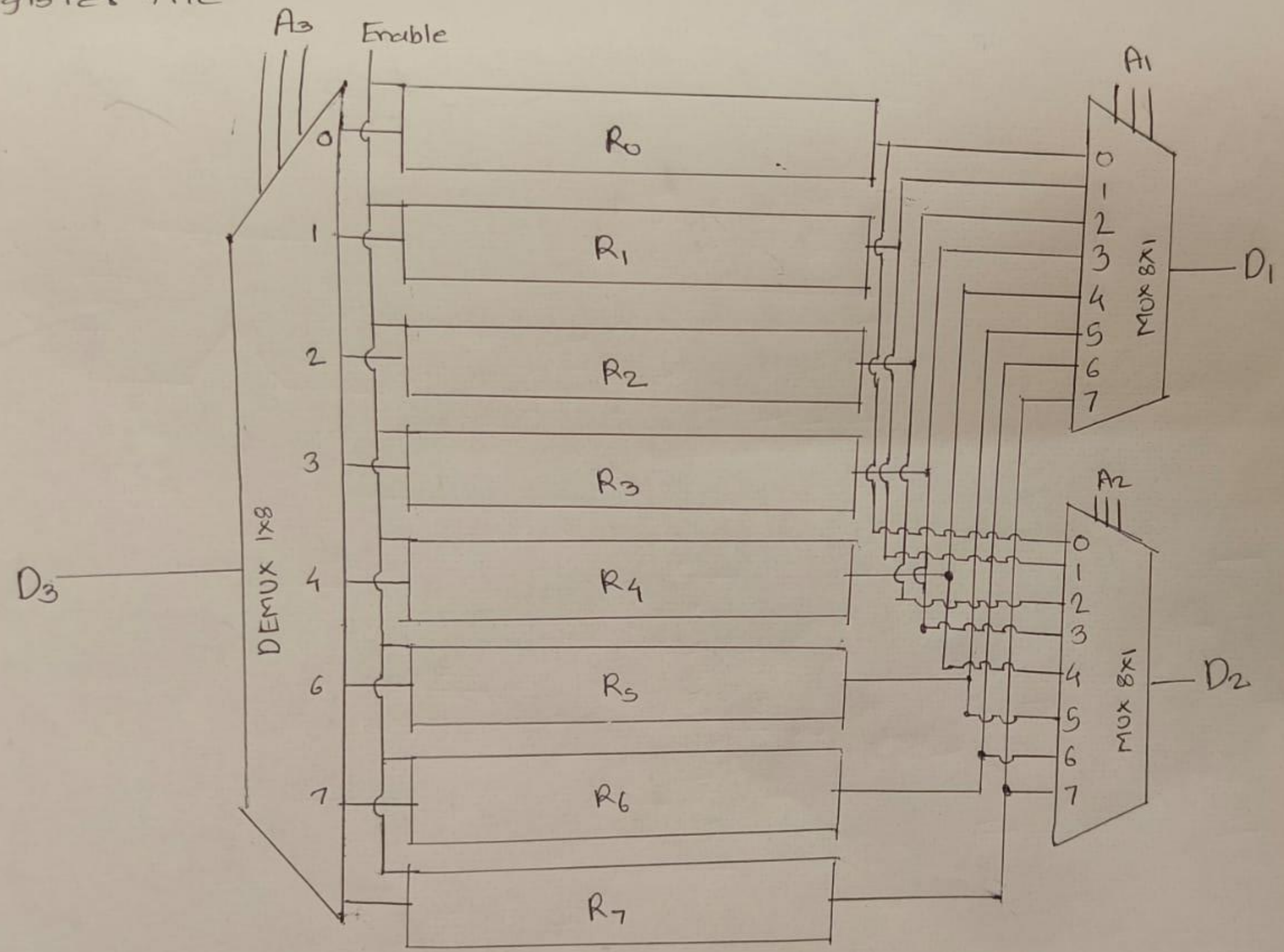


S0
Decoder-1
S15
S14
S1
Decoder-2
S11
S9
S8
S2
S13
S4
S3
Decoder-3
S10
S5
S7
S6

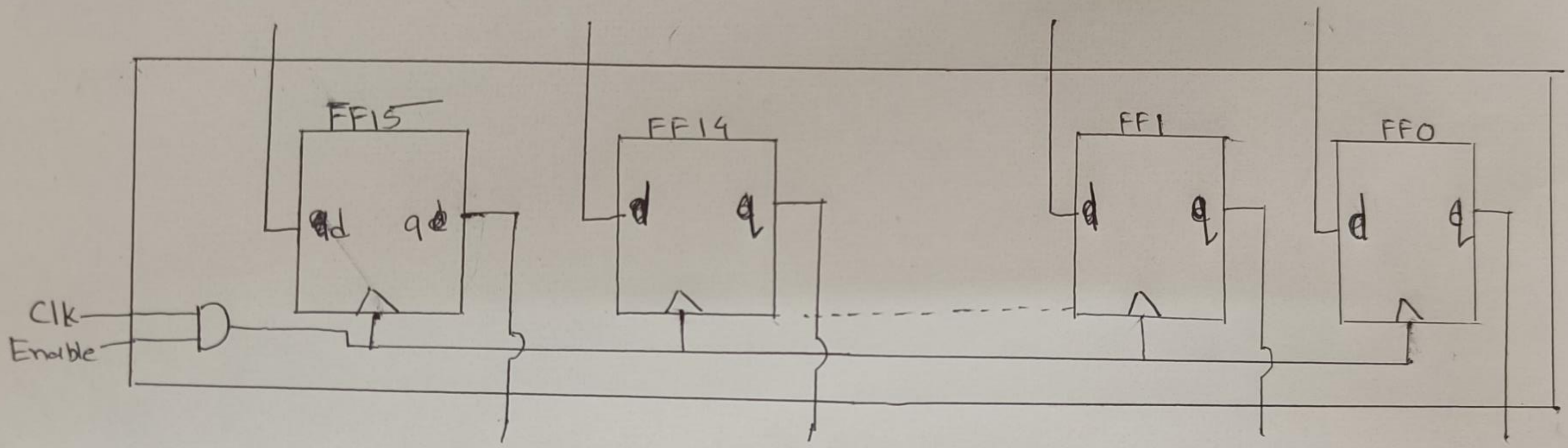


shift → 1s 8 times

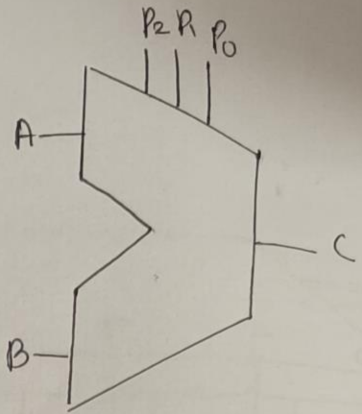
* Register file



* Register (16 bit)

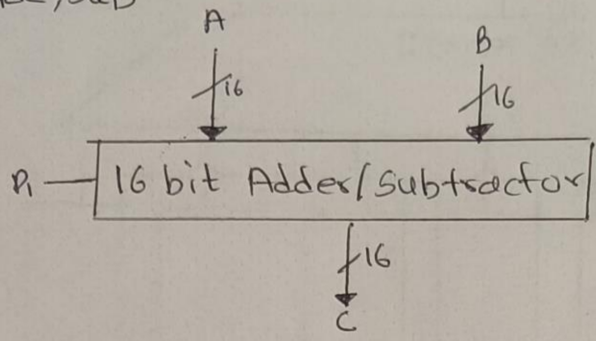


*ALU

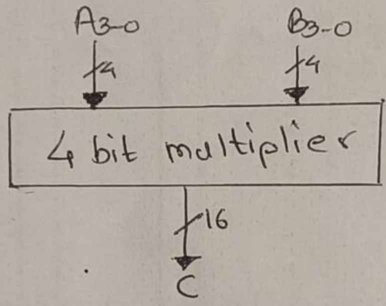


P_2	P_1	P_0	Operation
0	0	0	Add
0	0	1	—
0	1	0	SUB
0	1	1	MUL
1	0	0	AND
1	0	1	OR
1	1	0	IMP
1	1	1	—

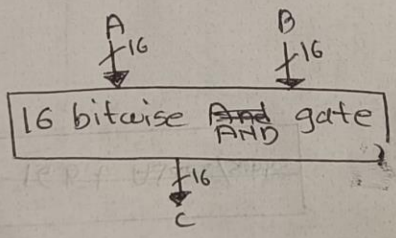
- Add, Sub



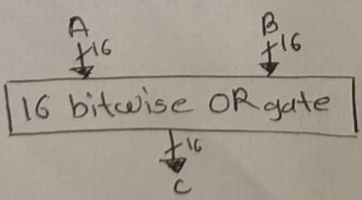
- Multiplication



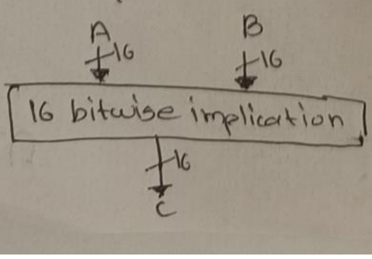
- Bitwise AND gate



- Bitwise OR gate



- Bitwise Implication ($A \to B$)



CONTROL SIGNALS

	P_C_en	IR_en	T_1_en	T2_en	RF_en	Mem_Read	Mem_Write	M_0 M_1	M_2	M_3	M_4	M_5 M_6	M_7 M_8	M_9 M_{10}	M_{11}
S_0	1	1				1		0 0	1	0	0				
S_1			1	1								1 0			
S_2			1					0 1	0			0 0			
S_3					1								1 1	0 1	
S_4			1					1 0	0						1
S_5			1			1					1	0 1			
S_6					1								1 1	0 0	
S_7							1				1				
S_8			1					0 1	0			0 0			
S_9	1							1 0	1	0					0
S_{10}					1								1 1	1 0	
S_{11}					1								1 0	0 0	
S_{12}	1									1					
S_{13}	1							1 1	1	0					
S_{14}					1								0 0	0 0	
S_{15}					1								0 1	0 0	

THANK YOU